

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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1. Experiment Title:

Palindrome Checker — a web page using HTML, CSS and JavaScript that checks whether a user-entered string is a palindrome, demonstrating string manipulation, closures, regular expressions, and exception handling (try/catch/throw).

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, JavaScript

3. Experiment Program Code:

[index.html](#)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>PALINDROME CHECKER</title>
  <link rel="stylesheet" href="styles.css">
  <script src="palindrome.js"></script>
</head>
<body>
  <div class="container">
    <h2>PALINDROME CHECKER</h2>
    <div class="card">
      <label for="word">Enter a String:</label><br><br>
      <input type="text" id="word" class="form-control" placeholder="e.g. racecar"><br><br>
      <input type="button" class="btn" value="Check Palindrome"
onclick="checkPalindrome()">
    </div>
  </div>
</body>
</html>
```

[styles.css](#)

```
/* Simple student project CSS for Palindrome Checker */
body {
  font-family: Arial, sans-serif;
  background-color: #f4f6f9;
  color: #333;
  margin: 0;
  padding: 20px;
}

.container {
  max-width: 450px;
  margin: 40px auto;
  background-color: #fff;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
}

h2 {
```

```

    text-align: center;
    color: #2c3e50;
    margin-top: 0;
    margin-bottom: 20px;
}

/* Card Styling */
.card {
    border: 1px solid #ddd;
    border-radius: 6px;
    padding: 25px;
    background-color: #fafafa;
}

label {
    font-weight: bold;
    color: #34495e;
}

.form-control {
    width: 100%;
    padding: 10px;
    border: 1px solid #ccc;
    border-radius: 4px;
    box-sizing: border-box;
    font-size: 15px;
}

.form-control:focus {
    outline: none;
    border-color: #3498db;
}

.btn {
    display: inline-block;
    width: 100%;
    padding: 10px;
    background-color: #3498db;
    color: #fff;
    border: none;
    border-radius: 4px;
    font-size: 16px;
    cursor: pointer;
    font-weight: bold;
    transition: background-color 0.2s;
}

.btn:hover {
    background-color: #2980b9;
}

```

[*palindrome.js*](#)

```

function reverseString(str){
    return str.split("").reverse().join("");
}

function message(){
    var msg = "Palindrome Checker";
    function display(){
        return msg;
    }
    return display;
}

function checkPalindrome(){
    try{
        var word = document.getElementById("word").value;
        if(word == "")
        {

```

```

        throw "Please enter a string";
    }
    if(!/^[A-Za-z0-9]+$/.test(word)){
        throw "Only alphabets/digits are allowed!";
    }
    if(word.length<3){
        throw "Please enter atleast 3 characters";
    }
    if(word.length>30){
        throw "Maximum 30 characters allowed";
    }
    let input = word.toLowerCase();
    var reverse = reverseString(input);
    if(input==reverse){
        document.write("<h2>" + message()() + "</h2>");
        document.write("<hr>");
        document.write("<b> Word:</b>" + word + "<br><br>");
        document.write("<b>Result: </b> PALINDROME");
    }
    else{
        document.write("<h2>" + message()() + "</h2>");
        document.write("<hr>");
        document.write("<b> Word:</b>" + word + "<br><br>");
        document.write("<b>Result: </b> NOT A PALINDROME");
    }
}
catch(error){
    alert(error);
}
}

```

4. Output:

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical4/index.html

PALINDROME CHECKER

Enter a String:

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical4/index.html — Input: "racecar"

Palindrome Checker

Word: racecar

Result: PALINDROME

Palindrome Checker

Word:hello

Result: NOT A PALINDROME

5. Result/Conclusion:

A Palindrome Checker web page was successfully designed and implemented using HTML, CSS and JavaScript. The program validates the input using a regular expression and length checks inside a try/catch block, reverses the string using a closure-returning helper function, and compares it with the original to display whether the entered word is a palindrome. This experiment reinforced the concepts of string manipulation, closures, regular expressions, and exception handling in JavaScript.

Upload the Soft copy as well as complete code on GitHub link is mandatory.

File name: Practical 4_24070521038_Vansh Mahant_A_A1